Advanced Econometrics – Exam

August 29, 2022

Exercise 1 (40%)

Quoting the magazine the Economist "...Unilever [is] one of the world's largest consumer-goods firms. Since a merger of British soapmakers and Dutch margarine merchants in 1929, Unilever has been a dual-nationality company. It is legally domiciled in Britain and the Netherlands, with headquarters in both the London building and in Rotterdam. [...] Under its current structure, the shares of Unilever NV cannot be exchanged for those in Unilever PLC, its British sibling." The Figure 1 presents quotations of the Unilever stocks on both stock exchanges.

Figure 1: Unilever quotations on UK and NL stock exchanges

Use 5% significance level. Using the figure and the output below answer and explain following questions.

a) (10%) Decide and explain integration orders of the two variables.

b) (10%) Are these two variables cointegrated? Explain.

c) (10%) What is the cointegrating vector? Interpret its elements.

d) (10%) Does the Error Correction Mechanism work? Explain.

```
testdf2(variable = unilever$nl, test.type = "c",
+         max.augmentations = 4, max.order = 3)
  augmentations         adf     p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -0.16886037 0.9362843               4403     12.8746093526926
2             1 -0.05626952 0.9511087               4402      0.0209819696639
3             2  0.01634953 0.9569725               4401      0.0000009142996
4             3  0.00635893 0.9561658               4400      0.0000208056759
5             4 -0.02039561 0.9540054               4399      0.0005938343952
  bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2
1        19.6694296290         19.708448562   0.0003330702  0.00005355964
2         6.7711776192          6.912935281   0.8848280855  0.03385770074
3         0.0013171310          0.028708287   0.9992370708  0.99934165130
4         0.0013238053          0.002005851   0.9963606027  0.99933831634
5         0.0009735681          0.005818210   0.9805585007  0.99951333439
  p_bg.p.value.3
1   0.0001950708
2   0.0747252075
3   0.9987173964
4   0.9999761216
5   0.9998821730
```

```
testdf2(variable = unilever$uk, test.type = "c",
+         max.augmentations = 4, max.order = 3)
  augmentations         adf     p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -0.12320535 0.9426183               4341        0.23340173525
2             1 -0.13764092 0.9406155               4340        0.00015022485
3             2 -0.09841285 0.9460580               4339        0.00002639909
4             3 -0.09565157 0.9464411               4338        0.00006955839
5             4 -0.19781908 0.9322664               4337        0.00007727632
  bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2
1         3.2856959474          3.310316413      0.6290129      0.1934284
2         3.0463477789          3.084881906      0.9902209      0.2180188
3         0.0092114441          0.041902723      0.9959005      0.9954049
4         0.0091124533          0.009611831      0.9933456      0.9954541
5         0.0005805892          0.010446056      0.9929861      0.9997097
  p_bg.p.value.3
1      0.3462094
2      0.3787223
3      0.9977472
4      0.9997501
5      0.9997169
```

```
unilever$dnl <- diff.xts(unilever$nl)
unilever$duk <- diff.xts(unilever$uk)
testdf2(variable = unilever$dnl, test.type = "nc",
+         max.augmentations = 4, max.order = 3)
  augmentations       adf p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -69.56396  0.01               4329       0.020705638162
2             1 -49.34342  0.01               4328       0.001182714346
3             2 -39.54130  0.01               4327       0.000594910943
4             3 -33.50393  0.01               4326       0.000003403441
5             4 -30.84891  0.01               4325       0.001136525400
  bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2
1         4.6889120292          4.688978262      0.8855837     0.09589936
2         0.0013679152          0.038708676      0.9725657     0.99931628
3         0.0006306439          0.003900113      0.9805409     0.99968473
4         0.0005452664          0.009219768      0.9985280     0.99972740
5         0.0017750759          0.002334789      0.9731065     0.99911286
  p_bg.p.value.3
1      0.1960407
2      0.9979979
3      0.9999353
4      0.9997652
5      0.9999700


testdf2(variable = unilever$duk, test.type = "nc",
+         max.augmentations = 4, max.order = 3)
  augmentations       adf p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -67.18228  0.01               4329          0.007427765
2             1 -48.17405  0.01               4328          0.001213917
3             2 -38.42097  0.01               4327          0.008480806
4             3 -31.26277  0.01               4326          0.001017273
5             4 -28.30080  0.01               4325          0.001954259
  bgodfrey.statistic.2 bgodfrey.statistic.3 p_bg.p.value.1 p_bg.p.value.2
1          3.039294714           3.64222973      0.9313198      0.2187890
2          0.004926611           0.54106688      0.9722063      0.9975397
3          0.010865091           0.01135323      0.9266255      0.9945822
4          0.004846706           0.01844621      0.9745560      0.9975796
5          0.005669202           0.01904810      0.9647394      0.9971694
  p_bg.p.value.3
1      0.3027785
2      0.9097809
3      0.9996794
4      0.9993374
5      0.9993048
```

```
# Estimating cointegrating vector
model.coint <- lm(nl~uk, data = unilever)
Call:
lm(formula = nl ~ uk, data = unilever)
Coefficients:
               Estimate  Std. Error t value            Pr(>|t|)
(Intercept) 10.50630745  0.09551688   110.0 <0.0000000000000002 ***
uk           0.01077713  0.00005318   202.7 <0.0000000000000002 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.181 on 4329 degrees of freedom
Multiple R-squared:  0.9046,    Adjusted R-squared:  0.9046
F-statistic: 4.107e+04 on 1 and 4329 DF,  p-value: < 0.00000000000000022

testdf2(variable = residuals(model.coint), test.type="nc",
+         max.augmentations = 4, max.order = 4)
  augmentations       adf p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -2.889228  0.01               4330        0.70167041578
2             1 -2.917281  0.01               4329        0.00010684327
3             2 -2.866769  0.01               4328        0.00053334319
4             3 -2.802749  0.01               4327        0.00006163864
5             4 -2.786426  0.01               4326        0.00019026624
  bgodfrey.statistic.2 bgodfrey.statistic.3 bgodfrey.statistic.4 p_bg.p.value.1
1         1.6094538662          3.579952937           3.68351408      0.4022230
2         0.8639675166          2.839545876           2.92914198      0.9917528
3         0.0006733378          2.052273569           2.14672066      0.9815751
4         0.0034059825          0.003927495           0.06349808      0.9937359
5         0.0027837538          0.002998929           0.01257301      0.9889946
  p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1      0.4472100      0.3105396      0.4505282
2      0.6492199      0.4170313      0.5697525
3      0.9996634      0.5616283      0.7087945
4      0.9982985      0.9999346      0.9995065
5      0.9986091      0.9999564      0.9999803

# Estimating ECM
model.ecm <- lm(dnl~duk + lresid - 1, data = unilever)
Call:
lm(formula = dnl ~ duk + lresid - 1, data = unilever)
Coefficients:
         Estimate Std. Error t value             Pr(>|t|)
duk     0.0100414  0.0001601  62.735 < 0.0000000000000002 ***
lresid -0.0035975  0.0012762  -2.819              0.00484 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2671 on 4328 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.4765,    Adjusted R-squared:  0.4762
F-statistic:  1969 on 2 and 4328 DF,  p-value: < 0.00000000000000022
```

Exercise 2 (30%)
A health econometrician estimated a model for a dummy dependent variable. The variable of interest is `smoker`, and it =1 for current smokers, =0 for non-smokers. The econometrician decided to use the `sex` variable that codes sex of respondents (=1 for women, =0 for men), `age` - age in years and `bmi` - body-mass-index as covariates.

The econometrician obtained the output presented below. The output comes from R program and it is divided into 4 sections.

a) Is the model in section (a) a probit model or a logit model? Explain.

b) Interpret parameters of the model presented in section (a).

c) What hypothesis was tested in section (b)? What is the result and finding?

d) Interpret marginal effects presented in section (c).

e) Interpret appropriate R^2 statistics in section (d).

```
# -------------------------------------------------------------
# a)
Call:
glm(formula = smoker ~ sex + age + bmi, family = binomial(link = "probit"),
    data = smoke)

Deviance Residuals:
   Min       1Q  Median       3Q      Max
-1.966  -1.167   0.678   1.113   1.286

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.406732   0.857333   2.807 0.004997 **
sex         -0.801884   0.238325  -3.365 0.000766 ***
age         -0.004879   0.006890  -0.708 0.478877
bmi         -0.023148   0.029138  -0.794 0.426930
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 233.18  on 171  degrees of freedom
Residual deviance: 219.54  on 168  degrees of freedom
AIC: 227.54

Number of Fisher Scoring iterations: 4

# -------------------------------------------------------------
# b)
> probit_restricted = glm(smoker~sex, data=smoke,
+                           family=binomial(link="probit"))
> lrtest(probit, probit_restricted)
Likelihood ratio test

Model 1: smoker ~ sex + age + bmi
```

```
Model 2: smoker ~ sex
  #Df  LogLik Df Chisq Pr(>Chisq)
1   4 -109.77
2   2 -110.50 -2 1.454     0.4834

# ---------------------------------------------------------------
# c)
 # marginal effects for the average observation
> (meff = probitmfx(formula=smoker~sex+age+bmi,
+                   data=smoke, atmean=TRUE))
Call:
probitmfx(formula = smoker ~ sex + age + bmi, data = smoke, atmean = TRUE)

Marginal Effects:
         dF/dx  Std. Err.       z     P>|z|
sex -0.2868432  0.0752012 -3.8143 0.0001365 ***
age -0.0018909  0.0026701 -0.7082 0.4788254
bmi -0.0089720  0.0112921 -0.7945 0.4268798
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

dF/dx is for discrete change for the following variables:
[1] "sex"

> # mean marginal effects
> (meff = probitmfx(formula=smoker~sex+age+bmi,
+                   data=smoke, atmean=FALSE))
Call:
probitmfx(formula = smoker ~ sex + age + bmi, data = smoke, atmean = FALSE)

Marginal Effects:
         dF/dx  Std. Err.       z    P>|z|
sex -0.2852462  0.0747266 -3.8172 0.000135 ***
age -0.0017840  0.0025085 -0.7112 0.476972
bmi -0.0084647  0.0105966 -0.7988 0.424400
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

dF/dx is for discrete change for the following variables:
[1] "sex"

# ---------------------------------------------------------------
# d)
> options(scipen=999)
> PseudoR2(probit)
        McFadden     Adj.McFadden        Cox.Snell       Nagelkerke McKelvey.Zavoina
      0.05850302       0.01561829       0.07624972       0.10272955       0.12241859
          Effron            Count        Adj.Count              AIC    Corrected.AIC
      0.07937974       0.61046512       0.05633803     227.54129117     227.78081213
```

Exercise 3 (30%)

The problem set uses data on choice of heating system in California hospitals. The observations consist of hospitals in California that were newly built and had central air-conditioning. The choice is among heating systems. Five types of systems are considered to have been possible:

- gas central (gc),
- gas room (gr),
- electric central (ec),
- electric room (er),
- heat pump (hp).

There are 900 observations with the following variables:

- idcase gives the observation number (1-900),
- depvar identifies the chosen alternative (gc, gr, ec, er, hp),
- ic.alt is the installation cost for the 5 alternatives,
- oc.alt is the annual operating cost for the 5 alternatives,
- income is the annual income of the hospital,
- agehed is the age of the hospital director,
- rooms is the number of rooms in the hospital,
- region a factor with levels ncostl (northern coastal region), scostl (southern coastal region), mountn (mountain region), valley (central valley region).

Note that the attributes of the alternatives, namely, installation cost and operating cost, take a different value for each alternative. Therefore, there are 5 installation costs (one for each of the 5 systems) and 5 operating costs. To estimate the logit model, the researcher needs data on the attributes of all the alternatives, not just the attributes for the chosen alternative. For example, it is not sufficient for the researcher to determine how much was paid for the system that was actually installed (ie., the bill for the installation). The researcher needs to determine how much it would have cost to install each of the systems if they had been installed. The importance of costs in the choice process (i.e., the coefficients of installation and operating costs) is determined through comparison of the costs of the chosen system with the costs of the non-chosen systems.

A health economist decided to estimate a model for the choice of heating system.

a) Was it a multinomial logit or conditional logit? Explain.

b) Interpret parameters.

c) What other variables could be used to model this dependent variable.

d) Name diagnostic tests that should be performed for the presented model.

```
Call:
mlogit(formula = depvar ~ ic + oc | income + agehed, data = H,
    method = "nr")

Frequencies of alternatives:choice
      ec       er       gc       gr       hp
0.071111 0.093333 0.636667 0.143333 0.055556

nr method
6 iterations, 0h:0m:0s
g'(-H)^-1g = 2.55E-05
successive function values within tolerance limits

Coefficients :
                  Estimate  Std. Error z-value    Pr(>|z|)
(Intercept):er  1.42227244  0.72737919  1.9553     0.05054 .
(Intercept):gc  0.28510088  0.73730020  0.3867     0.69899
(Intercept):gr -0.78700163  0.83130981 -0.9467     0.34379
(Intercept):hp -1.15034093  0.91609784 -1.2557     0.20923
ic             -0.00152607  0.00062384 -2.4463     0.01443 *
oc             -0.00699265  0.00155701 -4.4911 0.000007087 ***
income:er      -0.03863382  0.09960043 -0.3879     0.69810
income:gc      -0.00885934  0.07889200 -0.1123     0.91059
income:gr      -0.11580564  0.09141831 -1.2668     0.20524
income:hp       0.05986710  0.11350156  0.5275     0.59788
agehed:er      -0.02512549  0.01202438 -2.0895     0.03666 *
agehed:gc      -0.00432124  0.00944276 -0.4576     0.64722
agehed:gr      -0.00083136  0.01097521 -0.0757     0.93962
agehed:hp      -0.01878033  0.01355995 -1.3850     0.16606
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -1001.5
McFadden R^2:  0.020282
Likelihood ratio test : chisq = 41.466 (p.value = 0.0000093317)
>
```